

Requirement Understanding

Project: Community Library Membership & Book Lending System

1. Project Overview

The goal of this project is to build a Console-based Library Management System using .NET Core, EF Core, and PostgreSQL.

The system manages members, books, borrowing, returns, fines, and reports.

The main focus is on implementing proper business rules, database relationships, validations, and transaction handling.

2. Architecture & Design Understanding

I planned the project using layered architecture to keep the code clean and maintainable.

- Core Layer → Models and Enums
- DAL Layer → DbContext and Repository classes
- BLL Layer → Business logic and validations
- Presentation Layer → Console menus and user interaction

I used interfaces to separate implementation and improve flexibility.

3. Main Features Planned

Member Management

- Add and view members
- Search by email or phone
- Update membership status
- Deactivate members

Book Management

- Add books and categories
- Add multiple book copies
- Search books
- Mark copies as damaged/unavailable

Borrowing Module

- Validate borrowing rules
- Check membership limit
- Check the unpaid fine limit
- Prevent duplicate active borrowing
- Use EF Core transaction for a safe borrowing process

Return & Fine Module

- Return borrowed books
- Calculate late fine (₹10/day)
- Update copy availability
- Store return details

Fine Management

- View pending fines
- Pay fines
- View fine history

4. Database & Relationship Understanding

Main entities identified:

- Member
- MembershipType
- Book
- BookCategory
- BookCopy
- Borrowing
- FinePayment

Relationships were designed using primary keys and foreign keys.

- One book → many copies
- One member → many borrowings
- One category → many books

5. PostgreSQL Function Requirement

I understood that at least one PostgreSQL function must be created and called from EF Core.

Planned function:

- `calculate_member_fine(member_id)`
- Get the total unpaid fine before borrowing validation.

6. Reports Planned

The system includes reports such as:

- Borrowed books
- Overdue books
- Members with pending fines
- Most borrowed books
- Available books by category
- Member borrowing history

7. Validation

The system should handle validations for:

- Invalid member
- Invalid book
- No available copy
- Fine limit exceeded
- Borrowing limit exceeded
- Duplicate borrowing

ER diagram

